SEFI to Quantum Field Theory Mapping

Jason Dutton

Overview

This document outlines a basic mapping between SEFI parameters and standard Quantum Field Theory (QFT) quantities.

Frequency and Energy

$$\omega_c \leftrightarrow \frac{E}{\hbar}$$

Wave Number and Momentum

$$k \leftrightarrow \frac{p}{\hbar}$$

Field Mapping

$$\Psi \leftrightarrow \psi$$

Interpretation

SEFI curvature and resonance parameters can be related to QFT energy, momentum, and field amplitudes, providing a bridge between geometric and field-theoretic descriptions.